Learning to Lens

An Introduction to General Relativity and Strong Gravitational Lensing

with Mathematica

Based on the published works of

P. Schneider, J. Ehlers & E. E. Falco, *Gravitational Lenses* (1992)

R. Narayan & M. Bartelmann, *Lectures on Gravitational Lensing* (1997)

A. O. Petters, H. Levine & J. Wambsganss, *Singularity Theory and Gravitational Lensing* (2001)

S. Carroll, *Spacetime and Geometry* (2004)

A. B. Congdon & C. R. Keeton, *Principles of Gravitational Lensing* (2018)

M. Meneghetti, *Introduction to Gravitational Lensing* (2021)

P. Saha, D. Sluse, J. Wagner & L. L. R. Williams, *Essentials of Strong Gravitational Lensing* (2024)

— Instructor Edition (with Solutions) —

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March 19, 2026

About This Document

This tutorial suite was produced by **Rodrigo Córdova Rosado** (Harvard–Smithsonian Center for Astrophysics) with the assistance of **Claude** (Anthropic), an AI language model, via **Claude Code** — Anthropic's command-line tool for collaborative software engineering and technical writing.

The physics content in these notes draws from the following published textbooks and review articles:

- S. Carroll, *Spacetime and Geometry: An Introduction to General Relativity* (Addison Wesley, 2004)
- A. B. Congdon & C. R. Keeton, *Principles of Gravitational Lensing* (Springer, 2018)
- R. Narayan & M. Bartelmann, *Lectures on Gravitational Lensing* (1997; arXiv:astro-ph/9606001)
- P. Saha, D. Sluse, J. Wagner & L. L. R. Williams, *Essentials of Strong Gravitational Lensing*, Space Science Reviews **220**, 12 (2024)
- M. Meneghetti, *Introduction to Gravitational Lensing: With Python Examples* (Springer, 2021)
- P. Schneider, J. Ehlers & E. E. Falco, *Gravitational Lenses* (Springer, 1992)
- A. O. Petters, H. Levine & J. Wambsganss, *Singularity Theory and Gravitational Lensing* (Birkhäuser, 2001)

The workflow was as follows: the author directed Claude to co-write the LaTeX lecture notes, Wolfram Mathematica scripts, and interactive notebooks, drawing on the reference texts above. Claude drafted the text and code; the author reviewed, revised, and approved all content.

Importantly, all mathematical derivations and symbolic results in these notes have been verified deterministically using Wolfram Mathematica's computer algebra system — not generated probabilistically by an LLM. Every equation, identity, and worked solution was checked by running the companion `.wl` scripts (via `wolframscript`), which perform exact symbolic computation. The Mathematica verification outputs are reproducible and accompany each module in the repository.

The AI assistant was used for:

- Drafting expository prose and pedagogical explanations
- Structuring the curriculum and organizing modules
- Writing Mathematica code (subsequently executed and verified)
- Formatting LaTeX and managing cross-references

The AI assistant was **not** used as a source of mathematical truth. All physics content derives from the published works listed above, and all symbolic computations are verified by Mathematica.

This is a **living document**. Additional details, examples, and lessons will be added over time as the author continues to learn and teach this material. The most current version is always available at: https://github.com/rodelcr/Learning_to_Lens

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Notation Guide

Symbol	Meaning	Units / Notes
Coordinates and Angles		
$\boldsymbol{\theta}$	Image position (angular)	radians or arcsec
$\boldsymbol{\beta}$	Source position (angular)	radians or arcsec
$\boldsymbol{\alpha}$	Reduced deflection angle	radians
$\hat{\alpha}$	Physical deflection angle	radians
$\boldsymbol{\xi}$	Position in lens plane	physical (kpc, Mpc)
Distances		
D_{d}	Observer–lens angular diameter distance	Mpc
D_{s}	Observer–source angular diameter distance	Mpc
D_{ds}	Lens–source angular diameter distance	Mpc
Lensing Quantities		
θ_{E}	Einstein radius	arcsec
Σ_{cr}	Critical surface mass density	$\mathrm{g\,cm^{-2}}$ or $M_\odot\,\mathrm{pc}^{-2}$
κ	Convergence $= \Sigma/\Sigma_{\mathrm{cr}}$	dimensionless
γ	Shear magnitude	dimensionless
ψ	Effective lensing potential	dimensionless
μ	Magnification	dimensionless
$\mathcal{A}$	Jacobian (inverse magnification) matrix	2×2
Cosmology		
H_0	Hubble constant	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Ω_{m}	Matter density parameter	dimensionless
Ω_Λ	Dark energy density parameter	dimensionless
GR / Metric		
$g_{\mu\nu}$	Metric tensor	
$\Gamma^{\alpha}{}_{\mu\nu}$	Christoffel symbols	
$R^{\alpha}{}_{\beta\mu\nu}$	Riemann curvature tensor	
$R_{\mu\nu}$	Ricci tensor	
R	Ricci scalar	

Mathematica files are cross-linked throughout these notes. Links formatted as `filename.nb` or `filename.wl` point to companion Mathematica notebooks and scripts in the `Mathematica/`

directory of this repository.

Part I

General Relativity Foundations

Chapter 1

Special Relativity and Tensor Basics

> *General relativity (GR) is Einstein's theory of space, time, and gravitation. At heart it is a very simple subject . . . The essential idea is perfectly straightforward: while most forces of nature are represented by fields defined on spacetime, gravity is inherent in spacetime itself.* — Sean Carroll, *Spacetime and Geometry*

Before we can understand how gravity bends light, we need the language of special relativity and tensors. This chapter reviews the foundations: Minkowski spacetime, Lorentz transformations, four-vectors, and the index notation that will carry us through the rest of these notes. The emphasis is on building fluency with the notation and the key physical ideas — time dilation, length contraction, the invariant interval — so that the transition to curved spacetime in later modules feels natural.

Companion Mathematica files:

- `lorentz_transforms.wl` — symbolic construction and verification of Lorentz boost matrices
- `four_vectors.nb` — interactive visualizations of worldlines, light cones, and time dilation

1.1 From Newton to Einstein

In Newtonian mechanics, space and time are independent. The gravitational force between two masses M and m separated by a distance r is (cf. Carroll eq. 1.1)

$$\mathbf{F} = -\frac{GMm}{r^2}\,\hat{\mathbf{r}}\,, \tag{1.1}$$

and the gravitational potential Φ satisfies Poisson's equation,

$$\nabla^2\Phi = 4\pi G\rho\,. \tag{1.2}$$

In GR, both of these are replaced by statements about the curvature of spacetime. The key equation we will eventually arrive at is Einstein's field equation (Carroll eq. 1.5):

$$R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} = 8\pi G\,T_{\mu\nu}\,. \tag{1.3}$$

But we are getting ahead of ourselves. For now, we focus on *flat* spacetime — the arena of special relativity — and learn the tensor language needed to write down equations like (1.3).

1.2 Spacetime and the Invariant Interval

Special relativity (SR) unifies space and time into a single four-dimensional continuum called **Minkowski space**. An individual point in spacetime is an **event**, and the path of a particle through spacetime is its **worldline**.

Unlike Newtonian mechanics, SR has no absolute notion of simultaneity. Instead, the fundamental invariant is the **spacetime interval** between two events (Carroll eq. 1.10):

$$\boxed{(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \tag{1.4}$$

where we work in **natural units** with $c = 1$ throughout this chapter (we will restore factors of c when needed for astrophysical applications).

Remark 1.2.1. We adopt the **mostly-plus** sign convention $(-, +, +, +)$, following Carroll. Some references (especially in particle physics) use the opposite convention $(+, -, -, -)$. Be careful when comparing equations across textbooks.

The interval classifies the separation between events:

- $(\Delta s)^2 < 0$: **timelike** separation (events can be causally connected)
- $(\Delta s)^2 = 0$: **null** or **lightlike** separation (connected by a light ray)
- $(\Delta s)^2 > 0$: **spacelike** separation (no causal connection possible)

The **proper time** $\Delta\tau$ along a timelike path is defined as (Carroll eq. 1.17)

$$(\Delta\tau)^2 = -(\Delta s)^2 = (\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2 \,. \tag{1.5}$$

This is the time measured by a clock traveling along the worldline. For an infinitesimal displacement, we write the **line element** (Carroll eq. 1.20):

$$ds^2 = \eta_{\mu\nu}\, dx^\mu\, dx^\nu \,, \tag{1.6}$$

where we have introduced the **Minkowski metric** and the **Einstein summation convention** (repeated indices are summed over).

1.3 Index Notation and the Minkowski Metric

Spacetime coordinates are labeled by Greek indices running from 0 to 3 (Carroll eq. 1.12):

$$x^\mu : \quad x^0 = t, \quad x^1 = x, \quad x^2 = y, \quad x^3 = z \,. \tag{1.7}$$

Latin indices $i, j, k, \ldots$ run over the spatial components $1, 2, 3$ only.

The **Minkowski metric** is the 4×4 matrix (Carroll eq. 1.15):

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \,. \tag{1.8}$$

Its inverse, $\eta^{\mu\nu}$, has the same components. The metric is used to **raise and lower indices**:

$$V_\mu = \eta_{\mu\nu}\, V^\nu, \qquad \omega^\mu = \eta^{\mu\nu}\, \omega_\nu \,. \tag{1.9}$$

Note that in Minkowski space this simply flips the sign of the time component: $V_0 = -V^0$, while $V_i = V^i$.

1.4 Lorentz Transformations

The coordinate transformations that leave the interval (1.6) invariant are called **Lorentz transformations**. They act on the coordinates as (Carroll eq. 1.25)

$$x^{\mu'} = \Lambda^{\mu'}{}_{\nu}\, x^{\nu}\,, \tag{1.10}$$

where the matrix Λ satisfies the defining condition (Carroll eq. 1.28):

$$\boxed{\eta = \Lambda^{\mathrm{T}}\, \eta\, \Lambda} \tag{1.11}$$

or equivalently $\eta_{\rho\sigma} = \Lambda^{\mu'}{}_{\rho}\, \Lambda^{\nu'}{}_{\sigma}\, \eta_{\mu'\nu'}$.

1.4.1 Rotations

A spatial rotation in the x–y plane by angle θ is (Carroll eq. 1.31):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.12}$$

1.4.2 Boosts

A **Lorentz boost** along the x-axis with rapidity parameter ϕ is (Carroll eq. 1.32):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} \cosh\phi & -\sinh\phi & 0 & 0 \\ -\sinh\phi & \cosh\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.13}$$

Using $v = \tanh\phi$ and $\gamma = (1 - v^2)^{-1/2} = \cosh\phi$, we recover the familiar form (Carroll eq. 1.35):

$$\begin{aligned} t' &= \gamma(t - vx)\,, \\ x' &= \gamma(x - vt)\,. \end{aligned} \tag{1.14}$$

Exercise 1.4.1. Verify that the boost matrix (1.13) satisfies the Lorentz condition (1.11) by explicit matrix multiplication. Then confirm this symbolically using `lorentz_transforms.wl`.

1.4.3 Physical Consequences

The boost equations (1.14) immediately lead to the classic relativistic effects:

- **Time dilation:** A clock at rest in S' (i.e., $\Delta x' = 0$) ticks at intervals $\Delta t = \gamma\, \Delta t'$.
- **Length contraction:** A rod at rest in S' with proper length L_0 has measured length $L = L_0/\gamma$ in S.

- **Relativity of simultaneity:** Events simultaneous in S' ($\Delta t' = 0$) satisfy $\Delta t = \gamma v \Delta x$ in S.

Exercise 1.4.2. (Twin paradox.) An observer travels from event $A = (0, 0)$ to $B' = (\Delta t/2,\, v\Delta t/2)$ and back to $C = (\Delta t, 0)$ at constant speed v, while a stationary twin travels from A directly to C. Show that the traveling twin ages by $\Delta\tau_{AB'C} = \sqrt{1 - v^2}\,\Delta t$ while the stationary twin ages by $\Delta\tau_{ABC} = \Delta t$. (cf. Carroll eqs. 1.18–1.19. Visualize with `four_vectors.nb`.)

1.5 Four-Vectors

A **four-vector** V^μ is a set of four components that transforms under Lorentz transformations as (Carroll eq. 1.39):

$$V^{\mu'} = \Lambda^{\mu'}{}_{\nu}\, V^\nu\,. \tag{1.15}$$

Key examples:

Four-position: $x^\mu = (t, x, y, z)$.

Four-velocity: For a massive particle with worldline $x^\mu(\tau)$, the four-velocity is

$$U^\mu = \frac{dx^\mu}{d\tau} = \gamma(1, v^x, v^y, v^z)\,. \tag{1.16}$$

Its norm is always $\eta_{\mu\nu}U^\mu U^\nu = -1$ (in units with $c = 1$).

Four-momentum: $p^\mu = m\,U^\mu = (E, p^x, p^y, p^z)$, where $E = \gamma m$ is the relativistic energy and $\mathbf{p} = \gamma m\,\mathbf{v}$ is the three-momentum. The invariant norm gives the **energy–momentum relation**:

$$\boxed{\eta_{\mu\nu}\, p^\mu p^\nu = -E^2 + |\mathbf{p}|^2 = -m^2} \tag{1.17}$$

or, restoring c: $E^2 = (pc)^2 + (mc^2)^2$.

For a **photon** ($m = 0$), $E = |\mathbf{p}|$ and the four-momentum is null: $p^\mu p_\mu = 0$. This will be crucial when we trace light rays through curved spacetime in later modules.

1.6 Dual Vectors (One-Forms)

Associated with every vector space is a **dual vector space** (or cotangent space T_p^*). A dual vector (or **one-form**) ω_μ transforms under the *inverse* Lorentz transformation (Carroll eq. 1.50):

$$\omega_{\mu'} = \Lambda^\nu{}_{\mu'}\,\omega_\nu\,. \tag{1.18}$$

The prototypical one-form is the **gradient** of a scalar field ϕ (Carroll eq. 1.54):

$$(\mathrm{d}\phi)_\mu = \partial_\mu\phi \equiv \frac{\partial\phi}{\partial x^\mu}\,. \tag{1.19}$$

The contraction of a vector with a dual vector is a Lorentz scalar: $\omega_\mu V^\mu \in \mathbf{R}$.

1.7 Tensors

A **tensor** of type (k, l) is a multilinear map that takes k one-forms and l vectors and produces a real number (Carroll eq. 1.56). In components, the transformation law is (Carroll eq. 1.63):

$$T^{\mu'_1 \cdots \mu'_k}{}_{\nu'_1 \cdots \nu'_l} = \Lambda^{\mu'_1}{}_{\alpha_1} \cdots \Lambda^{\mu'_k}{}_{\alpha_k} \Lambda^{\beta_1}{}_{\nu'_1} \cdots \Lambda^{\beta_l}{}_{\nu'_l} T^{\alpha_1 \cdots \alpha_k}{}_{\beta_1 \cdots \beta_l} \,. \tag{1.20}$$

1.7.1 Important Operations

- **Contraction:** Summing over one upper and one lower index reduces the rank by $(1, 1)$ (Carroll eq. 1.70): $S^{\mu\rho}{}_{\sigma} = T^{\mu\nu\rho}{}_{\sigma\nu}$.
- **Raising/lowering:** Use the metric to move indices (Carroll eq. 1.72): $T^{\alpha\beta\mu}{}_{\delta} = \eta^{\mu\gamma} T^{\alpha\beta}{}_{\gamma\delta}$.
- **Symmetrization/antisymmetrization:** Round brackets denote symmetrization; square brackets denote antisymmetrization (Carroll eqs. 1.79–1.80).

1.7.2 Key Tensors in Minkowski Space

- **Metric:** $\eta_{\mu\nu}$ — a symmetric $(0, 2)$ tensor.
- **Kronecker delta:** $\delta^{\mu}{}_{\nu}$ — the identity map.
- **Levi-Civita symbol:** $\tilde{\epsilon}_{\mu\nu\rho\sigma}$ (Carroll eq. 1.68) — totally antisymmetric, with $\tilde{\epsilon}_{0123} = +1$.
- **Electromagnetic field strength:** $F_{\mu\nu} = -F_{\nu\mu}$ (Carroll eq. 1.69) — a $(0, 2)$ antisymmetric tensor encoding $\mathbf{E}$ and $\mathbf{B}$.

1.8 The Metric as the Central Object

The key insight that connects SR to GR is that the metric is the fundamental object describing the geometry of spacetime. In SR, the metric is the constant $\eta_{\mu\nu}$. In GR, the metric becomes a field $g_{\mu\nu}(x)$ that varies from point to point, encoding the curvature of spacetime due to matter and energy.

Everything we need for gravitational lensing — the bending of light, the distortion of images, the time delays between multiple images — follows from understanding how the metric determines the paths of light rays. That story begins in the next module, where we introduce curved spacetime and the geodesic equation.

1.9 Energy and Momentum in SR

The **stress-energy tensor** $T^{\mu\nu}$ encodes the energy and momentum content of matter and fields. For a perfect fluid with energy density ρ, pressure p, and four-velocity U^{μ}:

$$T^{\mu\nu} = (\rho + p)\, U^{\mu} U^{\nu} + p\, \eta^{\mu\nu} \,. \tag{1.21}$$

Conservation of energy and momentum is expressed as

$$\partial_\mu T^{\mu\nu} = 0\,. \tag{1.22}$$

This will generalize to $\nabla_\mu T^{\mu\nu} = 0$ in curved spacetime, and the stress-energy tensor will appear as the source term on the right-hand side of Einstein's field equation (1.3).

1.10 Summary

- Spacetime is a four-dimensional continuum; the invariant quantity is the interval $ds^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu$.
- Lorentz transformations ($x^{\mu'} = \Lambda^{\mu'}{}_\nu x^\nu$) leave the interval invariant. Boosts mix space and time.
- Four-vectors and tensors are classified by how they transform under Lorentz transformations.
- The metric $\eta_{\mu\nu}$ is the central object — it defines distances, raises/lowers indices, and generalizes to $g_{\mu\nu}$ in GR.
- Photons travel on null paths ($ds^2 = 0$), which will be the starting point for understanding gravitational lensing.

1.11 Problems

Exercise 1.11.1. Classify the following pairs of events as timelike, spacelike, or null separated:

(a) $A = (0, 0, 0, 0)$, $B = (2, 1, 1, 0)$

(b) $A = (0, 0, 0, 0)$, $B = (1, 1, 0, 0)$

(c) $A = (0, 0, 0, 0)$, $B = (3, 1, 2, 2)$

Check your answers with Mathematica.

Exercise 1.11.2. Show that two successive boosts along the x-axis with rapidities ϕ_1 and ϕ_2 are equivalent to a single boost with rapidity $\phi = \phi_1 + \phi_2$. Relate this to the velocity addition formula $v = (v_1 + v_2)/(1 + v_1 v_2)$.

Exercise 1.11.3. Prove that the four-velocity of any massive particle satisfies $U^\mu U_\mu = -1$ (in units with $c = 1$).

Exercise 1.11.4. A photon with energy E in frame S is observed in frame S' moving with velocity v along the x-axis.

(a) Show that the photon energy in S' is $E' = \gamma E(1 - v\cos\theta)$, where θ is the angle between the photon's direction and the x-axis in frame S.

(b) Specialize to $\theta = 0$ (head-on) and $\theta = \pi/2$ (transverse) and interpret physically.

Exercise 1.11.5. The electromagnetic field strength tensor has components (Carroll eq. 1.69):

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

(a) Show that $F_{\mu\nu}$ is antisymmetric.

(b) Compute $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$ and verify that the signs of the electric field components change.

(c) Compute the Lorentz scalar $F_{\mu\nu}F^{\mu\nu}$ and express it in terms of $\mathbf{E}$ and $\mathbf{B}$.

1.12 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_01a.wl`, which verifies each answer symbolically.

Exercise 1.11.1: Interval Classification

We compute $(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$:

(a) $(\Delta s)^2 = -(2)^2 + (1)^2 + (1)^2 + (0)^2 = -4 + 1 + 1 = -2 < 0 \Rightarrow$ **Timelike**. A massive particle can travel between these events.

(b) $(\Delta s)^2 = -(1)^2 + (1)^2 + (0)^2 + (0)^2 = 0 \Rightarrow$ **Null**. A light ray connects these events.

(c) $(\Delta s)^2 = -(3)^2 + (1)^2 + (2)^2 + (2)^2 = -9 + 1 + 4 + 4 = 0 \Rightarrow$ **Null**. Again lightlike: $|\Delta \mathbf{x}|^2 = 9 = (\Delta t)^2$.

Exercise 1.11.2: Boost Composition

The boost matrix for rapidity ϕ is $\Lambda(\phi) = \left(\begin{smallmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{smallmatrix}\right)$ (showing only the t–x block). Computing the product:

$$\begin{aligned}
\Lambda(\phi_1)\,\Lambda(\phi_2) &= \begin{pmatrix} \cosh\phi_1\cosh\phi_2 + \sinh\phi_1\sinh\phi_2 & -\cosh\phi_1\sinh\phi_2 - \sinh\phi_1\cosh\phi_2 \\ -\sinh\phi_1\cosh\phi_2 - \cosh\phi_1\sinh\phi_2 & \sinh\phi_1\sinh\phi_2 + \cosh\phi_1\cosh\phi_2 \end{pmatrix} \\
&= \begin{pmatrix} \cosh(\phi_1+\phi_2) & -\sinh(\phi_1+\phi_2) \\ -\sinh(\phi_1+\phi_2) & \cosh(\phi_1+\phi_2) \end{pmatrix} = \Lambda(\phi_1+\phi_2)\,,
\end{aligned}$$

using the hyperbolic addition formulas.

For the velocity addition formula, we use $v = \tanh\phi$:

$$v = \tanh(\phi_1+\phi_2) = \frac{\tanh\phi_1 + \tanh\phi_2}{1 + \tanh\phi_1\tanh\phi_2} = \frac{v_1 + v_2}{1 + v_1 v_2}\,.$$

Note: for $v_1 = v_2 = 1$ (speed of light), $v = 2/2 = 1$, confirming that the speed of light is invariant.

Exercise 1.11.3: Four-Velocity Norm

The four-velocity is $U^\mu = \gamma(1, v^1, v^2, v^3)$ where $\gamma = (1-v^2)^{-1/2}$ and $v^2 = (v^1)^2 + (v^2)^2 + (v^3)^2$. Then:

$$\begin{aligned}
U^\mu U_\mu = \eta_{\mu\nu} U^\mu U^\nu &= \gamma^2\left[-(1)^2 + (v^1)^2 + (v^2)^2 + (v^3)^2\right] \\
&= \gamma^2(-1 + v^2) = \frac{-(1-v^2)}{1-v^2} = \boxed{-1}\,.
\end{aligned}$$

Exercise 1.11.4: Photon Energy (Relativistic Doppler)

(a) A photon with energy E propagating at angle θ to the x-axis has four-momentum $p^\mu = E(1, \cos\theta, \sin\theta, 0)$. Applying the boost Λ along x:

$$E' = p^{0'} = \gamma(p^0 - v\,p^1) = \gamma E(1 - v\cos\theta)\,.$$

(b) Special cases:

- $\theta = 0$ (photon along $+x$, same direction as boost): $E' = \gamma E(1-v) = E\sqrt{\frac{1-v}{1+v}}$. This is a **redshift** ($E' < E$). The observer recedes from the source.
- $\theta = \pi/2$ (transverse): $E' = \gamma E = E/\sqrt{1-v^2}$. This is the **transverse Doppler effect** — a purely relativistic blueshift with no classical analogue, arising from time dilation.

Exercise 1.11.5: Electromagnetic Field Strength Tensor

(a) By inspection, $F_{\mu\nu} = -F_{\nu\mu}$ (the matrix is antisymmetric: the diagonal is zero, and off-diagonal elements differ by a sign).

(b) Raising both indices with the metric: $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$. In matrix form: $F^{\mu\nu} = \eta \cdot F_{\mu\nu} \cdot \eta$. The result is:

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & B_3 & -B_2 \\ -E_2 & -B_3 & 0 & B_1 \\ -E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

The electric field components flip sign (from raising the time index), while the magnetic components are unchanged.

(c) The Lorentz scalar:

$$\begin{aligned} F_{\mu\nu}F^{\mu\nu} &= 2\big(-E_1^2 - E_2^2 - E_3^2 + B_1^2 + B_2^2 + B_3^2\big) \\ &= 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)\,. \end{aligned}$$

This is a Lorentz invariant: all inertial observers agree on the value of $|\mathbf{B}|^2 - |\mathbf{E}|^2$.

Chapter 2

Differential Geometry and the Metric Tensor

> *The appropriate mathematical structure used to describe curvature is that of a differentiable manifold: essentially, a kind of set that looks locally like flat space, but might have a very different global geometry.* — Sean Carroll, *Spacetime and Geometry*

In Module 1a we worked exclusively in flat Minkowski spacetime. To describe gravity, we must allow spacetime to be *curved*. This chapter introduces the mathematical machinery of differential geometry — manifolds, the general metric tensor, the covariant derivative, Christoffel symbols, and the curvature tensors — that forms the foundation of general relativity and, ultimately, gravitational lensing.

Companion Mathematica files:

- `christoffel_symbols.wl` — general-purpose symbolic computation of Christoffel symbols and curvature tensors from any input metric
- `curved_surfaces.nb` — interactive visualizations of parallel transport, geodesics, and curvature on 2D surfaces

2.1 Manifolds and Coordinate Charts

A **manifold** is a space that may be curved globally but looks like $\mathbb{R}^n$ in every sufficiently small neighborhood. More precisely, an n-dimensional manifold M is a topological space equipped with a collection of **coordinate charts** — invertible maps $\phi : U \to \mathbb{R}^n$ from open subsets $U \subset M$ — that cover all of M and are smoothly compatible on overlaps (Carroll Sec. 2.2).

Examples of manifolds include:

- $\mathbb{R}^n$ (trivially)
- The 2-sphere S^2 (the surface of a ball)
- The 2-torus T^2 (the surface of a doughnut)
- Spacetime itself (a 4-dimensional Lorentzian manifold)

A key fact: in general, *no single coordinate system can cover an entire manifold*. The sphere S^2, for instance, requires at least two charts (e.g., stereographic projections from the north and south poles; Carroll eqs. 2.9–2.10). We will always work locally in a single chart, keeping in mind that our results are coordinate-independent.

2.2 The General Metric Tensor

On a manifold, we need a way to measure distances and angles. This is provided by the **metric tensor** $g_{\mu\nu}(x)$, a symmetric, nondegenerate $(0, 2)$ tensor field. The line element is (cf. Carroll eq. 1.20, now generalized):

$$\boxed{ds^2 = g_{\mu\nu}(x)\, dx^\mu\, dx^\nu} \tag{2.1}$$

In flat Minkowski space, $g_{\mu\nu} = \eta_{\mu\nu}$ everywhere. In curved spacetime, $g_{\mu\nu}$ varies from point to point and encodes the gravitational field.

The **inverse metric** $g^{\mu\nu}$ is defined by

$$g^{\mu\lambda} g_{\lambda\nu} = \delta^\mu{}_\nu\,. \tag{2.2}$$

As before, the metric raises and lowers indices: $V_\mu = g_{\mu\nu} V^\nu$, $V^\mu = g^{\mu\nu} V_\nu$.

Example 2.2.1. The metric on the 2-sphere of radius R in spherical coordinates (θ, ϕ) is

$$ds^2 = R^2\big(d\theta^2 + \sin^2\theta\, d\phi^2\big)\,, \tag{2.3}$$

so $g_{\theta\theta} = R^2$, $g_{\phi\phi} = R^2 \sin^2\theta$, and all other components vanish.

Example 2.2.2. The flat plane in polar coordinates (r, ϕ) has metric

$$ds^2 = dr^2 + r^2\, d\phi^2\,. \tag{2.4}$$

Despite the coordinate-dependent appearance, this space is flat — the curvature tensors vanish. We will verify this in Mathematica.

2.3 Vectors and Tensors on Manifolds

On a general manifold, vectors live in the **tangent space** T_p at each point p. A vector V at p is identified with a directional derivative operator along a curve $x^\mu(\lambda)$ through p (Carroll Sec. 2.3):

$$V = V^\mu \partial_\mu\,, \tag{2.5}$$

where $\partial_\mu \equiv \partial/\partial x^\mu$ form a **coordinate basis** for T_p.

Under a coordinate transformation $x^\mu \to x^{\mu'}$, vector components transform as (Carroll eq. 2.19):

$$V^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\nu}\, V^\nu\,. \tag{2.6}$$

This generalizes the Lorentz transformation law from Module 1a to arbitrary coordinate changes.

Dual vectors (one-forms) and general tensors transform analogously, with one factor of $\partial x^{\mu'}/\partial x^\nu$ for each upper index and $\partial x^\nu/\partial x^{\mu'}$ for each lower index.

2.4 The Covariant Derivative

Ordinary partial derivatives of tensors are *not* tensors on a curved manifold: the transformation of $\partial_\mu V^\nu$ picks up an extra, non-tensorial term from the coordinate dependence of the basis vectors (Carroll Sec. 3.2).

To fix this, we introduce the **covariant derivative** ∇_μ, which adds a correction term involving the **connection coefficients** $\Gamma^\rho{}_{\mu\sigma}$ (Carroll eq. 3.5):

$$\boxed{\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu{}_{\mu\lambda} V^\lambda} \tag{2.7}$$

For a one-form (Carroll eq. 3.16):

$$\nabla_\mu \omega_\nu = \partial_\mu \omega_\nu - \Gamma^\lambda{}_{\mu\nu} \omega_\lambda \,. \tag{2.8}$$

The general rule for a tensor of arbitrary rank: $+\Gamma$ for each upper index, $-\Gamma$ for each lower index (Carroll eq. 3.17).

The connection coefficients are *not* tensors — they transform inhomogeneously (Carroll eq. 3.10) — but the covariant derivative $\nabla_\mu V^\nu$ *is* a tensor.

2.5 Christoffel Symbols

In GR we demand two additional properties of the connection:

1. **Torsion-free:** $\Gamma^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\nu\mu}$
2. **Metric compatibility:** $\nabla_\rho g_{\mu\nu} = 0$

These uniquely determine the **Christoffel symbols** (also called the Levi-Civita connection; Carroll eq. 3.27):

$$\boxed{\Gamma^\sigma{}_{\mu\nu} = \frac{1}{2} g^{\sigma\rho}\big(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big)} \tag{2.9}$$

This is one of the most important formulas in GR. Given any metric, we can compute the Christoffel symbols and from them all of the geometry.

Example 2.5.1. For the plane in polar coordinates, $ds^2 = dr^2 + r^2 d\phi^2$ (Example 2.2.2), the only nonvanishing Christoffel symbols are (Carroll eqs. 3.29–3.31):

$$\Gamma^r{}_{\phi\phi} = -r \,, \qquad \Gamma^\phi{}_{r\phi} = \Gamma^\phi{}_{\phi r} = \frac{1}{r} \,. \tag{2.10}$$

We derive these and more general cases in `christoffel_symbols.wl`.

2.6 Parallel Transport

In flat space, we can freely slide a vector from one point to another while keeping it "constant." On a curved manifold, this is no longer unambiguous — the result depends on the *path taken*.

The notion of keeping a vector constant along a curve $x^\mu(\lambda)$ is formalized as **parallel transport**: a vector V^μ is parallel-transported if (Carroll eq. 3.40)

$$\frac{dV^\mu}{d\lambda} + \Gamma^\mu{}_{\rho\sigma}\frac{dx^\rho}{d\lambda}V^\sigma = 0\,. \tag{2.11}$$

An important consequence of metric compatibility is that parallel transport preserves the inner product: if V^μ and W^μ are both parallel-transported along a curve, then $g_{\mu\nu}V^\mu W^\nu = \text{const}$ (Carroll eq. 3.42).

On the 2-sphere, parallel transporting a vector around a closed loop results in a rotation of the vector — the angle of rotation is proportional to the enclosed area and the curvature. This is visualized interactively in `curved_surfaces.nb`.

2.7 The Geodesic Equation

A **geodesic** is the curved-space generalization of a straight line: it is the path that parallel-transports its own tangent vector. Setting $V^\mu = dx^\mu/d\lambda$ in (2.11), we obtain the **geodesic equation** (Carroll eq. 3.44):

$$\boxed{\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu{}_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0} \tag{2.12}$$

This can equivalently be derived by extremizing the proper time/interval functional (Carroll Sec. 3.3, eqs. 3.45–3.55; Congdon & Keeton eq. 3.60).

Key properties:

- **Timelike geodesics** ($ds^2 < 0$): paths of freely falling massive particles; parameterized by proper time τ.
- **Null geodesics** ($ds^2 = 0$): paths of light rays; parameterized by an affine parameter λ.
- **Spacelike geodesics** ($ds^2 > 0$): shortest spatial distances.
- The character (timelike/null/spacelike) is preserved along the geodesic (Carroll Sec. 3.4).

In flat space with Cartesian coordinates, $\Gamma^\mu{}_{\rho\sigma} = 0$, so the geodesic equation reduces to $d^2x^\mu/d\lambda^2 = 0$ — a straight line, as expected.

2.8 Curvature Tensors

The **Riemann curvature tensor** $R^\rho{}_{\sigma\mu\nu}$ measures how much parallel transport around a closed loop rotates a vector. It is defined by (Carroll eq. 3.4):

$$\boxed{R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho{}_{\nu\sigma} - \partial_\nu\Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}} \tag{2.13}$$

A manifold is flat if and only if $R^\rho{}_{\sigma\mu\nu} = 0$ everywhere. This is the mathematical statement that curvature is characterized entirely by the Riemann tensor.

From the Riemann tensor we construct:

- **Ricci tensor** (contraction on 1st and 3rd indices):

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu} \tag{2.14}$$

- **Ricci scalar** (trace of the Ricci tensor):

$$R = g^{\mu\nu} R_{\mu\nu} \tag{2.15}$$

These are the quantities that appear in Einstein's field equation

$$R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = 8\pi G\, T_{\mu\nu}\,, \tag{2.16}$$

relating the curvature of spacetime (left side) to its matter and energy content (right side).

Remark 2.8.1. The Riemann tensor in n dimensions has $n^2(n^2-1)/12$ independent components. In 4D spacetime, that is 20. In 2D, there is only 1 — the Gaussian curvature determines everything.

2.9 The Metric Determines Everything

Let us collect the chain of logic:

Metric $g_{\mu\nu}$
$\downarrow$ eq. (2.9)
Christoffel symbols $\Gamma^{\sigma}{}_{\mu\nu}$
$\downarrow$ eq. (2.12) $\downarrow$ eq. (2.13)
Geodesics (particle paths) **Curvature** $R^{\rho}{}_{\sigma\mu\nu}$

This is the central message: *the metric is the fundamental object in GR*. Once you have $g_{\mu\nu}$, you can compute everything — how particles move, how light bends, and how spacetime curves.

For gravitational lensing, the key application is: given the metric around a massive object (e.g., the Schwarzschild metric for a point mass), solve the null geodesic equation to trace light rays and determine how images are formed.

2.10 Summary

- A manifold is locally flat but may have global curvature; the metric $g_{\mu\nu}(x)$ encodes the geometry.
- The covariant derivative ∇_μ generalizes the partial derivative to curved spacetime, using the connection $\Gamma^{\sigma}{}_{\mu\nu}$.
- The Christoffel symbols are uniquely determined by the metric via (2.9) (torsion-free, metric-compatible).

- Parallel transport depends on the path; this path-dependence is measured by the Riemann curvature tensor.
- Geodesics — paths that parallel-transport their own tangent vector — are the trajectories of freely falling particles and light rays.
- Einstein's equation relates curvature ($R_{\mu\nu}$) to matter content ($T_{\mu\nu}$).

2.11 Problems

Exercise 2.11.1. Compute all nonvanishing Christoffel symbols for the 2-sphere metric $ds^2 = R^2(d\theta^2 + \sin^2\theta\, d\phi^2)$ by hand, then verify with `christoffel_symbols.wl`.

Exercise 2.11.2. For the flat plane in polar coordinates, $ds^2 = dr^2 + r^2 d\phi^2$:

(a) Compute the Christoffel symbols.

(b) Compute the Riemann tensor and verify that the space is flat ($R^\rho{}_{\sigma\mu\nu} = 0$).

(c) Write down the geodesic equations and show that straight lines in Cartesian coordinates ($x = a + b\lambda$, $y = c + d\lambda$) satisfy them.

Exercise 2.11.3. For the 2-sphere metric, compute:

(a) The Riemann tensor $R^\theta{}_{\phi\theta\phi}$.

(b) The Ricci tensor $R_{\theta\theta}$ and $R_{\phi\phi}$.

(c) The Ricci scalar R. Show that $R = 2/R^2$ (constant positive curvature, as expected for a sphere of radius R).

Exercise 2.11.4. A vector initially pointing east is parallel-transported from the equator of a unit sphere ($R = 1$) along the following path: north along the meridian $\phi = 0$ to the pole, then south along $\phi = \alpha$ back to the equator. Show that the vector has rotated by angle α. (Hint: solve the parallel transport equation (2.11) along each segment, or use the geometric argument that the rotation equals the solid angle enclosed.)

Exercise 2.11.5. Show that great circles are geodesics of the 2-sphere. Specifically, show that the equator $\theta = \pi/2$, $\phi = \lambda$ satisfies the geodesic equation (2.12) with the Christoffel symbols from Exercise 2.11.1.

2.12 Solutions to Exercises

Full worked solutions verified in `problems_01b.wl`.

Exercise 2.11.1: Christoffel Symbols of the 2-Sphere

For $ds^2 = R^2(d\theta^2 + \sin^2\theta\, d\phi^2)$, the metric is $g_{\theta\theta} = R^2$, $g_{\phi\phi} = R^2\sin^2\theta$, with $g^{\theta\theta} = 1/R^2$, $g^{\phi\phi} = 1/(R^2\sin^2\theta)$.

Applying (2.9):

$$\Gamma^{\theta}{}_{\phi\phi} = \frac{1}{2}g^{\theta\theta}(-\partial_\theta g_{\phi\phi}) = \frac{1}{2}\cdot\frac{1}{R^2}\cdot(-2R^2\sin\theta\cos\theta) = -\sin\theta\cos\theta$$
$$\Gamma^{\phi}{}_{\theta\phi} = \frac{1}{2}g^{\phi\phi}(\partial_\theta g_{\phi\phi}) = \frac{1}{2}\cdot\frac{1}{R^2\sin^2\theta}\cdot 2R^2\sin\theta\cos\theta = \cot\theta$$

All other components vanish (verified by Mathematica).

Exercise 2.11.2: Polar Coordinates

(a) For $ds^2 = dr^2 + r^2 d\phi^2$: $\Gamma^{r}{}_{\phi\phi} = -r$, $\quad \Gamma^{\phi}{}_{r\phi} = 1/r$.

(b) Computing the Riemann tensor from these Christoffel symbols: $R^{\rho}{}_{\sigma\mu\nu} = 0$ for all index combinations. The space is flat, confirming that the nonvanishing Christoffel symbols are purely a coordinate artifact.

(c) The geodesic equations are: $\ddot{r} - r\dot{\phi}^2 = 0$ and $\ddot{\phi} + (2/r)\dot{r}\dot{\phi} = 0$. A straight line $x = a + b\lambda$, $y = c + d\lambda$ in Cartesian coordinates satisfies these when converted to polar coordinates $r = \sqrt{x^2+y^2}$, $\phi = \arctan(y/x)$.

Exercise 2.11.3: Curvature of the 2-Sphere

(a) From (2.13): $R^{\theta}{}_{\phi\theta\phi} = \sin^2\theta$.

(b) Ricci tensor: $R_{\theta\theta} = R^{\lambda}{}_{\theta\lambda\theta} = R^{\phi}{}_{\theta\phi\theta} = 1$. $R_{\phi\phi} = R^{\theta}{}_{\phi\theta\phi} = \sin^2\theta$.

(c) Ricci scalar: $R = g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = \frac{1}{R^2}(1) + \frac{1}{R^2\sin^2\theta}(\sin^2\theta) = \frac{2}{R^2}$. This is constant and positive — the sphere has uniform positive curvature.

Exercise 2.11.4: Parallel Transport

Along a meridian ($\phi = \text{const}$), $d\phi/d\lambda = 0$, so the parallel transport equation reduces to $dV^\mu/d\lambda = 0$ — the vector components are constant along the path.

After traveling north to the pole along $\phi = 0$ and south along $\phi = \alpha$, the vector returns to the equator. However, the coordinate basis $\hat{\phi}$ at $\phi = \alpha$ is rotated by angle α relative to $\hat{\phi}$ at $\phi = 0$. The physical vector has therefore rotated by angle α.

Geometrically, the solid angle subtended by the triangular path (two meridians and a segment of the equator) is α. On a unit sphere, the holonomy angle equals the enclosed solid angle.

Exercise 2.11.5: Great Circles as Geodesics

For the equator: $\theta(\lambda) = \pi/2$, $\phi(\lambda) = \lambda$, so $\dot{\theta} = 0$, $\dot{\phi} = 1$, $\ddot{\theta} = 0$, $\ddot{\phi} = 0$.

θ-equation: $0 + \Gamma^{\theta}{}_{\phi\phi} \cdot 1 \cdot 1 = -\sin(\pi/2)\cos(\pi/2) = 0$. ✓

ϕ-equation: $0 + 2\Gamma^{\phi}{}_{\theta\phi} \cdot 0 \cdot 1 = 0$. ✓

By the rotational symmetry of the sphere, any great circle can be mapped to the equator by a rotation, so all great circles are geodesics.

Chapter 3

The Schwarzschild Solution

> *In GR, the unique spherically symmetric vacuum solution is the Schwarzschild metric; it is second only to Minkowski space in the list of important spacetimes.* — Sean Carroll, *Spacetime and Geometry*

Having developed the general machinery of differential geometry in Module 1b, we now apply it to the most important exact solution in GR: the **Schwarzschild metric**, which describes the spacetime geometry outside any spherically symmetric, non-rotating mass. This metric governs the gravitational field of stars, planets, and non-rotating black holes, and is the starting point for understanding gravitational lensing.

Companion Mathematica files:

- `schwarzschild_metric.wl` — derive and verify the Schwarzschild metric: compute Christoffel symbols, Riemann tensor, verify $R_{\mu\nu} = 0$, and compute the Kretschner scalar
- `gravitational_redshift.nb` — interactive plots of gravitational redshift, time dilation, and Flamm's paraboloid embedding diagram

3.1 The Most General Spherically Symmetric Metric

We seek the metric outside a static, spherically symmetric mass distribution. The most general line element consistent with these symmetries can be written (Carroll eq. 5.5; cf. Congdon & Keeton eq. 3.64):

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2\, d\Omega^2\,, \tag{3.1}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$ is the metric on the unit 2-sphere, and $\alpha(r)$ and $\beta(r)$ are functions to be determined from Einstein's equation.

Remark 3.1.1. The "static" assumption means the metric is independent of t and has no $dt\,dr$ cross terms. "Spherically symmetric" means the angular part takes the form $r^2 d\Omega^2$, with r chosen so that spheres at constant r have area $4\pi r^2$.

3.2 Deriving the Schwarzschild Metric

We solve Einstein's vacuum equation $R_{\mu\nu} = 0$ for the metric (3.1). The strategy follows Carroll Sec. 5.1:

Step 1: Christoffel symbols. The nonvanishing Christoffel symbols for (3.1) are (Carroll eq. 5.12):

$$\begin{aligned}
&\Gamma^t{}_{tr} = \alpha' \,, \quad \Gamma^r{}_{tt} = e^{2(\alpha-\beta)}\alpha' \,, \quad \Gamma^r{}_{rr} = \beta' \,, \\
&\Gamma^r{}_{\theta\theta} = -r\, e^{-2\beta} \,, \quad \Gamma^r{}_{\phi\phi} = -r\, e^{-2\beta} \sin^2\theta \,, \\
&\Gamma^\theta{}_{r\theta} = \Gamma^\phi{}_{r\phi} = \frac{1}{r} \,, \quad \Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta \,, \quad \Gamma^\phi{}_{\theta\phi} = \cot\theta \,,
\end{aligned} \tag{3.2}$$

where primes denote d/dr.

Step 2: Ricci tensor. Computing the Ricci tensor components (Carroll eq. 5.14):

$$R_{tt} = e^{2(\alpha-\beta)} \left[\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{2}{r}\alpha' \right] , \tag{3.3}$$

$$R_{rr} = -\alpha'' - (\alpha')^2 + \alpha'\beta' + \frac{2}{r}\beta' \,, \tag{3.4}$$

$$R_{\theta\theta} = e^{-2\beta}\big[r(\beta' - \alpha') - 1 \big] + 1 \,, \tag{3.5}$$

$$R_{\phi\phi} = R_{\theta\theta} \sin^2\theta \,. \tag{3.6}$$

Step 3: Solve $R_{\mu\nu} = 0$. From $e^{-2(\alpha-\beta)}R_{tt} + R_{rr} = 0$, we get $\frac{2}{r}(\alpha' + \beta') = 0$, hence (Carroll eq. 5.17):

$$\alpha = -\beta + \text{const} \,. \tag{3.7}$$

By rescaling the time coordinate, we set the constant to zero: $\alpha = -\beta$.

Setting $R_{\theta\theta} = 0$ (Carroll eq. 5.18):

$$e^{2\alpha}(1 + 2r\alpha') = 1 \quad \Longrightarrow \quad \partial_r(r\, e^{2\alpha}) = 1 \,. \tag{3.8}$$

Integrating:

$$e^{2\alpha} = 1 - \frac{R_S}{r} \,, \tag{3.9}$$

where R_S is an integration constant.

Step 4: Identify R_S. In the weak-field limit ($r \to \infty$), the metric must reduce to $g_{tt} \approx -(1 + 2\Phi/c^2)$ with $\Phi = -GM/r$. Comparing with $g_{tt} = -(1 - R_S/r)$, we identify the **Schwarzschild radius** (Carroll eq. 5.23):

$$\boxed{R_S = \frac{2GM}{c^2}} \tag{3.10}$$

Result: The Schwarzschild Metric.

$$\boxed{ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2} \tag{3.11}$$

This is the unique vacuum solution with spherical symmetry (**Birkhoff's theorem**; Carroll Sec. 5.2). We verify $R_{\mu\nu} = 0$ symbolically in `schwarzschild_metric.wl`.

3.3 Birkhoff's Theorem

Birkhoff's theorem states that the Schwarzschild metric is the *unique* spherically symmetric vacuum solution to Einstein's equation. The remarkable consequence: even a *time-dependent* spherically symmetric source (e.g., a pulsating star) produces a *static* exterior metric (Carroll Sec. 5.2).

This is the GR analogue of Newton's shell theorem: the gravitational field outside a spherically symmetric mass depends only on the total enclosed mass M, not on how the mass is distributed.

3.4 Gravitational Time Dilation

A clock at radial coordinate r in the Schwarzschild spacetime runs at a rate different from a clock at infinity. For a stationary clock ($dr = d\theta = d\phi = 0$), the proper time is related to coordinate time by (Congdon & Keeton eq. 3.65):

$$d\tau = \sqrt{1 - \frac{R_S}{r}}\, dt\,. \tag{3.12}$$

Since $\sqrt{1 - R_S/r} < 1$ for $r > R_S$, clocks closer to the mass run *slower* than clocks farther away. At $r = R_S$, the time dilation becomes infinite — this is the event horizon.

3.5 Gravitational Redshift

A photon emitted at radius $r_{\rm em}$ with frequency $\nu_{\rm em}$ is observed at $r_{\rm obs} \to \infty$ with frequency:

$$\frac{\nu_{\rm obs}}{\nu_{\rm em}} = \frac{\sqrt{1 - R_S/r_{\rm em}}}{\sqrt{1 - R_S/r_{\rm obs}}} \xrightarrow{r_{\rm obs}\to\infty} \sqrt{1 - \frac{R_S}{r_{\rm em}}}\,. \tag{3.13}$$

Since $\nu_{\rm obs} < \nu_{\rm em}$, the photon is **redshifted** as it climbs out of the gravitational potential well. The gravitational redshift parameter is:

$$z_{\rm grav} = \frac{\nu_{\rm em}}{\nu_{\rm obs}} - 1 = \left(1 - \frac{R_S}{r_{\rm em}}\right)^{-1/2} - 1\,. \tag{3.14}$$

In the weak-field limit ($R_S/r \ll 1$): $z_{\rm grav} \approx GM/(rc^2)$ (Congdon & Keeton eq. 3.63).

Remark 3.5.1. The gravitational redshift was one of the three classical tests of GR proposed by Einstein, along with the perihelion precession of Mercury and the bending of light. It has been measured to high precision using atomic clocks at different altitudes (e.g., Pound–Rebka experiment, 1959; GPS satellite corrections).

3.6 The Event Horizon and Singularities

The Schwarzschild metric has two apparent singularities: at $r = 0$ and at $r = R_S$. Their nature is quite different (Carroll Sec. 5.3):

$r = R_S$ **(coordinate singularity).** The metric components diverge, but this is a *coordinate artifact*. The curvature remains finite, and the geometry is perfectly smooth. In appropriate coordinates (e.g., Eddington–Finkelstein or Kruskal–Szekeres), the metric is regular at $r = R_S$. However, $r = R_S$ is the **event horizon**: once a particle crosses it, it can never return.

$r = 0$ **(true singularity).** This is a genuine physical singularity where the curvature becomes infinite. We can confirm this with a curvature invariant: the **Kretschner scalar**:

$$K = R_{\mu\nu\rho\sigma}\, R^{\mu\nu\rho\sigma} = \frac{12\, R_S^2}{r^6} = \frac{48\, G^2 M^2}{c^4\, r^6}\,. \tag{3.15}$$

This diverges as $r \to 0$ but is finite at $r = R_S$, confirming that only $r = 0$ is a true singularity. We verify eq. (3.15) in `schwarzschild_metric.wl`.

3.7 The Weak-Field Limit

For $r \gg R_S$, the Schwarzschild metric reduces to the **weak-field** form (Congdon & Keeton eq. 3.71):

$$ds^2 \approx -\left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 + \left(1 - \frac{2\Phi}{c^2}\right) \left[dr^2 + r^2\, d\Omega^2\right], \tag{3.16}$$

where $\Phi = -GM/r$ is the Newtonian gravitational potential. This form is valid for *any* weak gravitational field, not just the Schwarzschild case, and will be the starting point for deriving the deflection angle of light in Module 2.

Remark 3.7.1. The weak-field metric (3.16) has an important feature: the spatial part is modified by a factor $(1 - 2\Phi/c^2)$ as well as the time part. This means that gravity affects both the *rate* of clocks (time dilation) and the *curvature of space*. Both effects contribute equally to the bending of light — this is why GR gives twice the Newtonian deflection angle.

3.8 Schwarzschild Radius for Astrophysical Objects

Object	**Mass**	R_S	**Actual Radius**
Earth	6×10^{24} kg	8.9 mm	6371 km
Sun	2×10^{30} kg	3.0 km	7×10^5 km
Neutron star	$1.4\, M_\odot$	4.1 km	~ 10 km
SgrA*	$4 \times 10^6\, M_\odot$	1.2×10^7 km	— (black hole)
Galaxy ($10^{12}\, M_\odot$)	$10^{12}\, M_\odot$	3×10^{12} km	—

For a galaxy-mass lens ($M \sim 10^{12}\, M_\odot$), $R_S \sim 0.1$ pc — far smaller than the galaxy itself. This confirms that the weak-field limit is appropriate for gravitational lensing by galaxies.

3.9 Embedding Diagram: Flamm's Paraboloid

To visualize the spatial curvature of the Schwarzschild geometry, we can embed the equatorial ($\theta = \pi/2$) spatial slice in 3D Euclidean space. At constant time t, the spatial metric on the equatorial plane is:

$$dl^2 = \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\phi^2\,. \tag{3.17}$$

Embedding this as a surface of revolution $z(r)$ in cylindrical coordinates (r, ϕ, z) where $dl^2 = (1 + z'^2)dr^2 + r^2 d\phi^2$, we find:

$$z(r) = 2\sqrt{R_S}\,\sqrt{r - R_S}\,, \tag{3.18}$$

which is **Flamm's paraboloid**. This is plotted interactively in `gravitational_redshift.nb`.

3.10 Summary

- The Schwarzschild metric is the unique spherically symmetric vacuum solution (Birkhoff's theorem).
- It is parameterized by a single quantity: the mass M (or equivalently the Schwarzschild radius $R_S = 2GM/c^2$).
- Gravitational time dilation: clocks run slower in stronger gravitational fields; $d\tau = \sqrt{1 - R_S/r}\, dt$.
- Gravitational redshift: photons lose energy climbing out of a potential well.
- $r = R_S$ is a coordinate singularity (event horizon); $r = 0$ is a true curvature singularity.
- The weak-field limit ($ds^2 \approx -(1 + 2\Phi/c^2)c^2 dt^2 + (1 - 2\Phi/c^2)d\ell^2$) will be the foundation for light deflection in Module 2.

3.11 Problems

Exercise 3.11.1. Using the reusable functions from Module 1b (`christoffel_symbols.wl`), verify that the Schwarzschild metric (3.11) satisfies $R_{\mu\nu} = 0$.

Exercise 3.11.2. Compute the Kretschner scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ for the Schwarzschild metric and show that $K = 12\,R_S^2/r^6$. Verify that K is finite at $r = R_S$ but diverges at $r = 0$.

Exercise 3.11.3. Compute the Schwarzschild radius for:

(a) The Earth ($M = 5.97 \times 10^{24}$ kg)

(b) The Sun ($M = 1.99 \times 10^{30}$ kg)

(c) A typical galaxy lens ($M = 10^{12}\,M_\odot$)

In each case, compare R_S to the physical size of the object.

Exercise 3.11.4. A photon is emitted from the surface of a neutron star ($M = 1.4\,M_\odot$, $R = 10$ km).

(a) Compute the gravitational redshift z_{grav}.

(b) By what fraction is the photon's wavelength increased by the time it reaches a distant observer?

(c) Compare to the redshift from the surface of the Sun.

Exercise 3.11.5. Derive the Flamm's paraboloid embedding (3.18) by embedding the equatorial spatial slice of the Schwarzschild geometry in 3D Euclidean space with cylindrical coordinates (r, ϕ, z).

3.12 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_01c.wl`, which verifies each answer symbolically and numerically.

Exercise ??: Verify $R_{\mu\nu} = 0$ for Schwarzschild

The Schwarzschild metric in coordinates (t, r, θ, ϕ) is

$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - R_s/r} + r^2\, d\theta^2 + r^2 \sin^2\theta\, d\phi^2\,,$$

where $R_s = 2GM/c^2$. We work in geometric units $c = G = 1$ so that $R_s = 2M$.

Step 1: Christoffel symbols. The nonvanishing components are (using the symmetry $\Gamma^\sigma{}_{\mu\nu} = \Gamma^\sigma{}_{\nu\mu}$):

$$\Gamma^t{}_{tr} = \frac{R_s}{2r(r - R_s)}\,, \qquad \Gamma^r{}_{tt} = \frac{R_s(r - R_s)}{2r^3}\,,$$

$$\Gamma^r{}_{rr} = -\frac{R_s}{2r(r - R_s)}\,, \qquad \Gamma^r{}_{\theta\theta} = -(r - R_s)\,,$$

$$\Gamma^r{}_{\phi\phi} = -(r - R_s)\sin^2\theta\,, \qquad \Gamma^\theta{}_{r\theta} = \frac{1}{r}\,,$$

$$\Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta\,, \qquad \Gamma^\phi{}_{r\phi} = \frac{1}{r}\,,$$

$$\Gamma^\phi{}_{\theta\phi} = \cot\theta\,.$$

Step 2: Riemann and Ricci tensors. Substituting into the definition of $R^\rho{}_{\sigma\mu\nu}$ and contracting to form $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$, every component of $R_{\mu\nu}$ vanishes identically. This confirms that the Schwarzschild metric is a vacuum solution to Einstein's field equations ($G_{\mu\nu} = 0$).

Verified symbolically by Mathematica — see `problems_01c.wl`, Exercise 1.1.

Exercise 3.11.2: Kretschner Scalar

The Kretschner scalar is the full contraction of the Riemann tensor:

$$K = R^{\alpha\beta\gamma\delta}\, R_{\alpha\beta\gamma\delta}\,.$$

Computing all components of $R_{\alpha\beta\gamma\delta}$ (with lowered first index) and fully contracting, the result simplifies to

$$\boxed{K = \frac{12\, R_s^2}{r^6} = \frac{48\, M^2}{r^6}}\,.$$

Verified symbolically by Mathematica — see `problems_01c.wl`, Exercise 1.2.

Behavior at the horizon ($r = R_s$):

$$K(r = R_s) = \frac{12}{R_s^4}\,,$$

which is *finite*. The apparent singularity in the metric at $r = R_s$ is merely a coordinate singularity (removable by switching to, e.g., Eddington–Finkelstein coordinates).

Behavior at $r = 0$:

$$K \to \infty \quad \text{as } r \to 0\,.$$

This is a *true curvature singularity*: tidal forces diverge regardless of the coordinate system chosen.

Exercise 3.11.3: Schwarzschild Radii for Astrophysical Objects

Using $R_s = 2GM/c^2$ with $G = 6.674 \times 10^{-11}$ m^{3}kg^{-1}s^{-2} and $c = 2.998 \times 10^8$ m/s:

(a) **Earth** ($M = 5.972 \times 10^{24}$ kg, $R = 6371$ km):

$$R_s = \frac{2 \times 6.674 \times 10^{-11} \times 5.972 \times 10^{24}}{(2.998 \times 10^8)^2} \approx 8.87 \times 10^{-3} \text{ m} \approx 8.87 \text{ mm}\,.$$

The ratio $R_s/R \approx 1.4 \times 10^{-9}$: GR corrections on Earth's surface are of order 10^{-9} (important for GPS, but tiny).

(b) **Sun** ($M = 1.989 \times 10^{30}$ kg, $R = 6.957 \times 10^8$ m):

$$R_s \approx 2953 \text{ m} \approx 2.95 \text{ km}\,.$$

The ratio $R_s/R \approx 4.24 \times 10^{-6}$: small, but measurable via gravitational redshift and light deflection.

(c) **Galaxy lens** ($M \sim 10^{12}\, M_\odot$, extent ~ 50 kpc):

$$R_s \approx 2.95 \times 10^{15} \text{ m} \approx 0.096 \text{ pc}\,.$$

The ratio $R_s/R \approx 1.9 \times 10^{-6}$: the Schwarzschild radius is vastly smaller than the galaxy, justifying the weak-field/thin-lens approximation used in gravitational lensing.

Exercise ??: Gravitational Redshift from a Neutron Star

Parameters: $M = 1.4\, M_\odot = 2.785 \times 10^{30}$ kg, $R = 10$ km $= 10^4$ m.

First, the Schwarzschild radius:

$$R_s = \frac{2GM}{c^2} = \frac{2 \times 6.674 \times 10^{-11} \times 2.785 \times 10^{30}}{(2.998 \times 10^8)^2} \approx 4137 \text{ m} \approx 4.14 \text{ km}\,.$$

So $R_s/R \approx 0.414$ — a large fraction, meaning strong-field GR is essential.

(a) Gravitational redshift:

$$z = \frac{1}{\sqrt{1 - R_s/R}} - 1 = \frac{1}{\sqrt{1 - 0.414}} - 1 = \frac{1}{\sqrt{0.586}} - 1 \approx \frac{1}{0.7655} - 1 \approx 0.306\,.$$

(b) Wavelength shift: For a photon emitted at $\lambda_{\rm em} = 500$ nm:

$$\lambda_{\rm obs} = \lambda_{\rm em}(1+z) = 500 \times 1.306 \approx 653 \text{ nm}\,,$$

a shift of $\Delta\lambda \approx 153$ nm — blue/green light is shifted to red/near-infrared.

(c) Comparison to the Sun: The Sun's surface redshift is

$$z_\odot = \frac{1}{\sqrt{1 - R_s^\odot/R_\odot}} - 1 \approx \frac{R_s^\odot}{2R_\odot} \approx 2.12 \times 10^{-6}\,.$$

The neutron star redshift ($z \approx 0.306$) is roughly 1.4×10^5 times larger — a dramatic difference that strongly affects observed neutron star spectra.

Exercise ??: Flamm's Paraboloid

We embed the equatorial spatial slice ($t = \text{const}$, $\theta = \pi/2$) of the Schwarzschild geometry into flat 3D Euclidean space.

The 2D slice:

$$dl^2 = \frac{dr^2}{1 - R_s/r} + r^2\, d\phi^2\,.$$

3D Euclidean space in cylindrical coordinates:

$$dl^2_{\rm flat} = dz^2 + dr^2 + r^2\, d\phi^2\,.$$

For a surface of revolution $z = z(r)$, the induced metric on the surface is

$$dl^2 = \left(1 + \left(\frac{dz}{dr}\right)^2\right) dr^2 + r^2\, d\phi^2\,.$$

Matching the dr^2 components:

$$1 + \left(\frac{dz}{dr}\right)^2 = \frac{1}{1 - R_s/r} \quad \Longrightarrow \quad \left(\frac{dz}{dr}\right)^2 = \frac{R_s/r}{1 - R_s/r} = \frac{R_s}{r - R_s}\,.$$

Integrating:

$$z(r) = \int \sqrt{\frac{R_s}{r - R_s}}\, dr = 2\sqrt{R_s}\,\sqrt{r - R_s} = 2\sqrt{R_s(r - R_s)}\,,$$

where we set $z(R_s) = 0$ at the throat.

This is the equation of a **paraboloid of revolution** ($z^2 = 4R_s(r - R_s)$). Key features:

- The minimum circumferential radius is $r = R_s$ (the throat), where $z = 0$.
- At large r, $z \sim 2\sqrt{R_s\, r}$ and the surface becomes approximately flat.
- The diagram illustrates spatial curvature, not gravitational potential.
- The full paraboloid (including $z < 0$) represents an Einstein–Rosen bridge connecting two asymptotically flat regions.

Verified symbolically by Mathematica — see `problems_01c.wl`, Exercise 1.5.

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